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MIL-B-70824A (AR)
2 June 1994
SUPERSEDING
MIL-B-70824 (AR)
09 September 1988

MILITARY SPECIFICATION

BATTERY, LITHIUM THIONYL CHLORIDE, SINGLE CELL RESERVE

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements and quality assurance provisions for the fabrication of parts, assembling, and packing for a reserve Lithium Thionyl Chloride battery.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specifications for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6135

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MIL-T-63420 - Thionyl Chloride Electrolyte

STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-202	-	Electronic and Electrical Component Parts, Test Method for
MIL-STD-331	-	Fuze and Fuze Components Environmental and Performance Test for
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.5)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

Product and Packing Drawings:

12550985	-	Battery Assembly, Single Cell Reserve
12550986	-	Insulator, Disk
12550987	-	Pad, Ampule Support
12550988	-	Shim, Ampule Support
12550989	-	Dry Room Control
12550995	-	Terminal Plate Assembly
9275597	-	Glass
12550996	-	Plate, Terminal
12550998	-	Pin, Terminal
12551015	-	Ampule Assembly, Electrolyte
12551016	-	Blank, Ampule
9317641	-	Ampule Blank (Anneal)
9328788	-	Tube, Glass
12551020	-	Cell Assembly
12551025	-	Case Assembly, Insulator

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9275592 - Case
12551007 - Insulator, Bottom, Inside
12551027 - Insulator, Bottom, Outside
12551028 - Separator, Bottom
12551029 - Cell Assembly, Anode/Cathode
12551030 - Anode Assembly
12551031 - Pad, Anode
12551032 - Substrate, Anode, Formed
12551033 - Substrate, Anode
12551034 - Paper, Glass Fiber
12551035 - Cathode Assembly
12551036 - Substrate, Cathode
12551037 - Composition Cathode
12551039 - Vertical Separator
12550752 - Rise Time, Capacity, Internal Impedance Test Procedure for M762/M767 Reserve Cell Primer Activation
12550994 - Paper Glass Fiber
12550757 - Ballistic Environment Test Requirement for M762/M767 Components

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-E-493 - Standard test method of leaks

(Application for copies of ASTM should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria).

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3. REQUIREMENTS

3.1. Materials. Materials shall be in accordance with applicable drawings and specifications.

3.1.1 Components and assemblies. The Components and assemblies shall comply with all requirements specified on Drawing 12550985, all associated drawings, and with all requirements specified in applicable specifications and standards.

3.2 Temperature range.

3.2.1 Storage temperature range. Batteries manufactured to this specification shall be unaffected by storage temperatures of -60 degrees F to +160 degrees F.

3.2.2 Operating temperature range. Batteries manufactured to this specification shall be capable of meeting full performance requirements from -25 degrees F to +110 degrees F. Degraded performance for certain parameters is permitted from -60 degrees F to -25 degrees F and from +110 degrees F to +160 degrees F. The parameters for degraded performance are shown in the applicable portions of this specification.

3.3 Samples.

3.3.1 First article. When specified in the contract or purchases order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.3.2 Supplemental samples.

3.3.2.1 Change of qualified product. After qualification, the manufacturer shall notify the purchaser prior to the implementation of any change in design, material, or process which may affect performance, quality, reliability of interchangeability. Such notification shall include a thorough description of the proposed change and a suggested test plan designed to demonstrate that the change will not adversely affect performance, quality, reliability or interchangeability and that the changed product will continue to meet the specification requirements. The purchaser will review the proposed change and test plan and advise the manufacturer of the testing required to obtain approval of the proposed change. Upon completion of the prescribed tests, the results shall be submitted to the purchaser for review. Based upon the review,

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the purchaser will approve or disapprove the change and notify the manufacturer. At the manufacturer's option, devices incorporating the change may be manufactured and tested at the manufacturer's expense prior to approval; however, all shipments of these changed devices shall be withheld until approval is granted by the purchaser.

3.4 Electrical performance.

3.4.1 Non-activation voltage. Non-activated batteries built to this specification shall have a voltage of less than 5.0 millivolts, in accordance with 4.6.1.1.

3.4.2 Activation and rise time.

3.4.2.1 Activation. Battery activation shall be per procedure 12550752, "Rise Time, Capacity, Internal Impedance Test Procedure for M762/M767 Reserve Cell - Primer Activation."

3.4.2.2 Rise time. The rise time of batteries manufactured to this specification shall meet the following requirements:

Temperature	Maximum Time to 2.3 Volts
+110 degrees F	300 milliseconds
Ambient	800 milliseconds
-25 degrees F	1500 milliseconds
-60 degrees F	2500 milliseconds

3.4.3 Capacity rating. Batteries manufactured to this specification shall meet the following capacity requirements:

After 15 days with 13K Load:

Temperature	Minimum Voltage (RL = 13K)
+110 degrees F	3.50 volts
Ambient	3.40 volts
-25 degrees F	2.90 volts
-60 degrees F	2.40 volts

3.4.4 Internal impedance.

3.4.4.1 Internal impedance (Non-activated). The internal impedance of batteries manufactured to this specification shall be not less than 1 megohm. This requirement applies over the temperature range of -60 degrees F to +160 degrees F.

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3.4.4.2 Internal impedance (Activated). The internal impedance of batteries manufactured to this specification shall meet the following requirements after 15 days with a 13K load applied:

The minimum voltage with a 150 ohm load applied for 50ms shall be:

Temperature	Minimum Voltage
+110 degrees F	3.10 volts
Ambient	3.00 volts
-25 degrees F	2.20 volts
-60 degrees F	1.90 volts

3.5 Mechanical

3.5.1 Configuration. Configuration of the batteries manufactured to this specification shall be in accordance with assembly Drawing 12550985, Battery Assembly, Single Cell Reserve.

3.5.2 Leakage. All batteries manufactured to this specification shall be gross leak tested using a bubble test method. All batteries shall also be fine leak tested to 5 $X10^{-7}$ cc of Helium per second.

3.5.3. Ampule heat soak. The ampule assembly, electrolyte (12551015), shall withstand a 24 hour heat soak at 165 degrees F ± 5 degrees F without leakage or breakage, per 4.6.7.

3.6 Material and finish. The material and finish for batteries manufactured to this specification shall be in accordance with the Drawing 12550985. This shall include all requirements on sub-assembly and piece part drawings which are included in Drawing 12550985.

3.7 Environmental and ballistic tests.

3.7.1 Environmental tests (Pre-activation). Batteries produced to this specification shall not be activated by a 1.5 meter (5 foot) drop test as described in 4.6.5.1.

3.7.2 Ballistic tests (Post activation). Batteries procured to this specification shall exhibit continuous output for the first 50 milliseconds of flight when exposed to the following conditions when assembled in a M762 fuze:

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Set Back Acceleration	30,000 G's
Set Forward Shock	2,500 G's for 3 ms
Sideslap Shock	2,500 G's for 0.5 ms
Spin	30,000 RPM at center line, offset by 0.35 In.

3.7.3 Requalification. After successful completion of the first article acceptance test, the batteries are required to be tested for ballistic performance every six months, with the criteria for reintroduction of testing three consecutives lots in accordance with the test specified on Fig.2 ,when a lot fails the pass/fail criteria cited on 4.4.2.1. The number of batteries required for each temperature group shall be twelve.

3.8 Marking. Batteries shall be marked per 4.6.

3.9 Radiographic examination. Radiographic (x-ray film) batteries manufactured to this specification shall be performed in accordance with 4.6.2.

3.10 Workmanship. The batteries produced to this specification shall be fabricated and finished in a thorough workmanlike manner and all manufacturing and assembly operations shall be correctly performed. Metal parts and welds shall be free of burrs, chips, sharp edges, cracks, crazes, porosity, warpage, burn marks, chipped edges, dents, flash, dirt, grease, rust, corrosion products, deposits and all foreign matter or other defects.

3.10.1 Cleaning. After fabrication, the assembled battery shall be cleaned of smudges, loose or spattered excess weld metal, metal chips and burrs and any other foreign material which may detract from the intended operation, function or appearance of the battery. All corrosive material shall be removed. The cleaning method used shall not be injurious to any part of the assembly nor shall the part be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

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The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a First article inspection. (see 4.3)

b Quality conformance inspection (see 4.4)

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of two hundred and sixty batteries (260), plus ten (10) sets of each piece part and assembly listed on the Classification of Characteristics (C of C's) using the methods and processes established for production.

4.3.2 Inspection to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification (see MIL-A-48078 and TABLE I, Figures 1A and 1B) and be inspected for compliance with any or all requirements of the applicable drawings.

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TABLE I - First article inspection

CLASSIFICATION OF CHARACTERISTICS

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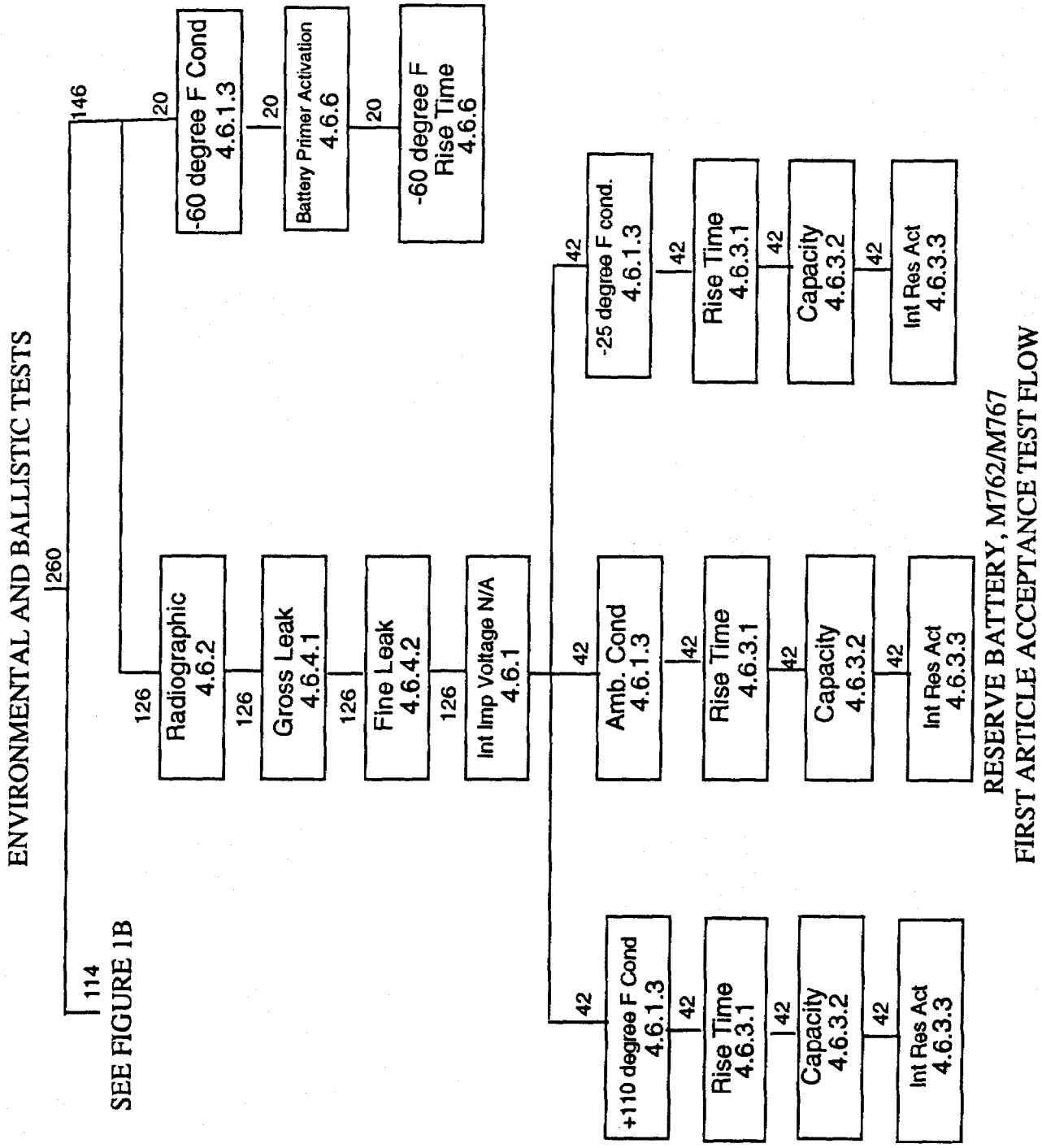
PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Battery, single cell, reserve	1 OF 1	12550985
			NEXT HIGHER ASSEMBLY: NONE

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Environmental and Ballistic Tests			
	Workmanship	100%	3.10	Visual
	Non-activation Voltage	126-0-1	3.4.1	4.6.1
	Gross Leak	126-0-1	3.5.2	4.6.4.1
	Fine Leak	126-0-1	3.5.2	4.6.4.2
	Five Foot Drop			
	Battery Primer Activation	20-0-1	3.4.2	4.6.6
	Risetime, Capacity, Internal Impedance Tests	126-0-1	3.4	4.6.3
	Ballistic (57 mm)	114-0-1	3.7.2	4.6.3.1
	Radiographic Test	126-0-1	3.9	4.6.2

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

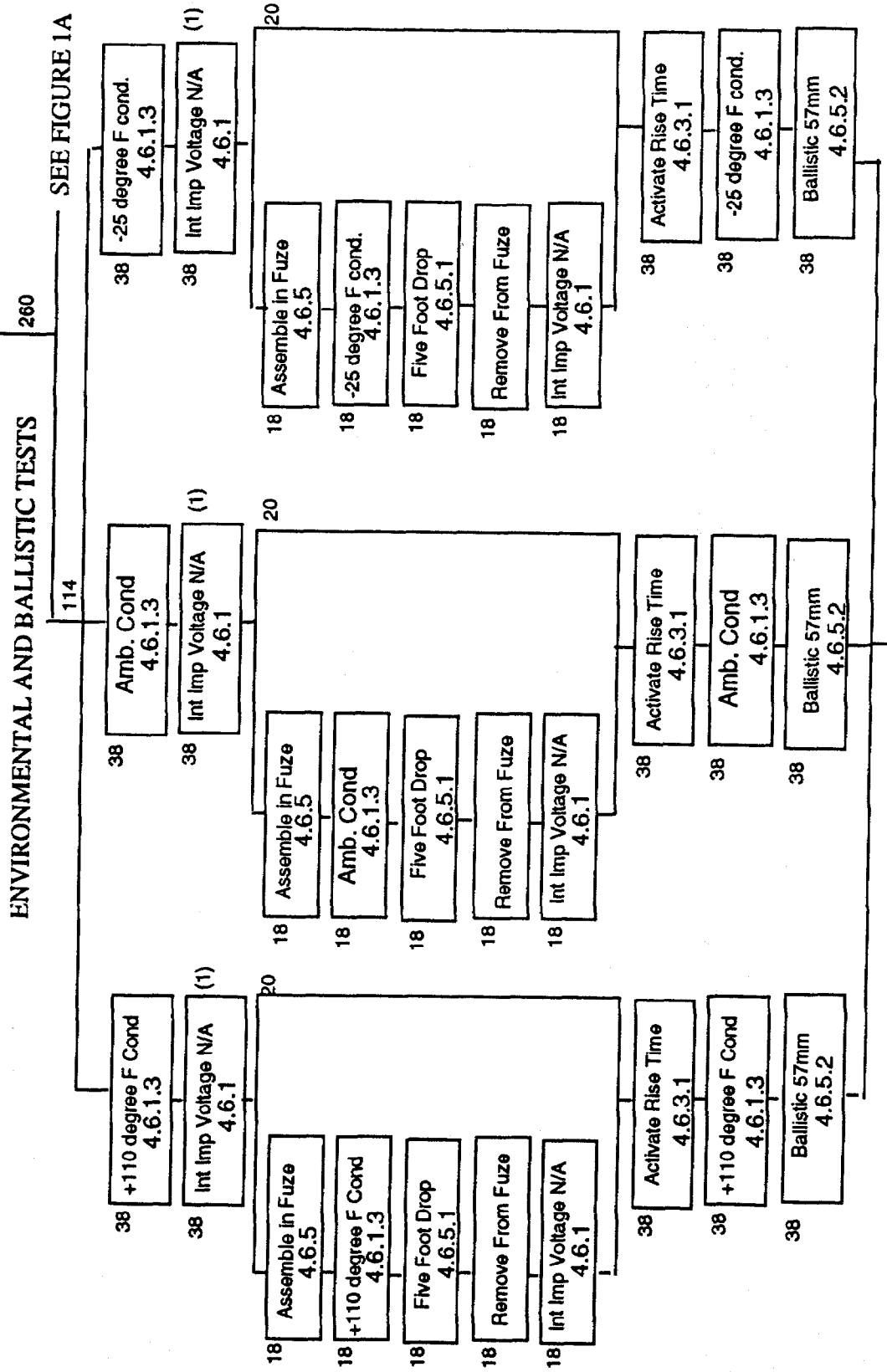
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RESERVE BATTERY, M762/M767
FIRST ARTICLE ACCEPTANCE TEST FLOW

FIGURE 1A

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NOTES:
(1) N/A - Not Activated
(2) Number of Batteries

RESERVE BATTERY, M762/M767
FIRST ARTICLE ACCEPTANCE TEST FLOW

FIGURE 1B

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4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of:

a. Sampling lot acceptance testing; and

b. Individual unit inspections

4.4.1 Sampling lot acceptance testing and individual unit inspection.

4.4.1.1 Lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078. In addition inspection lots shall contain:

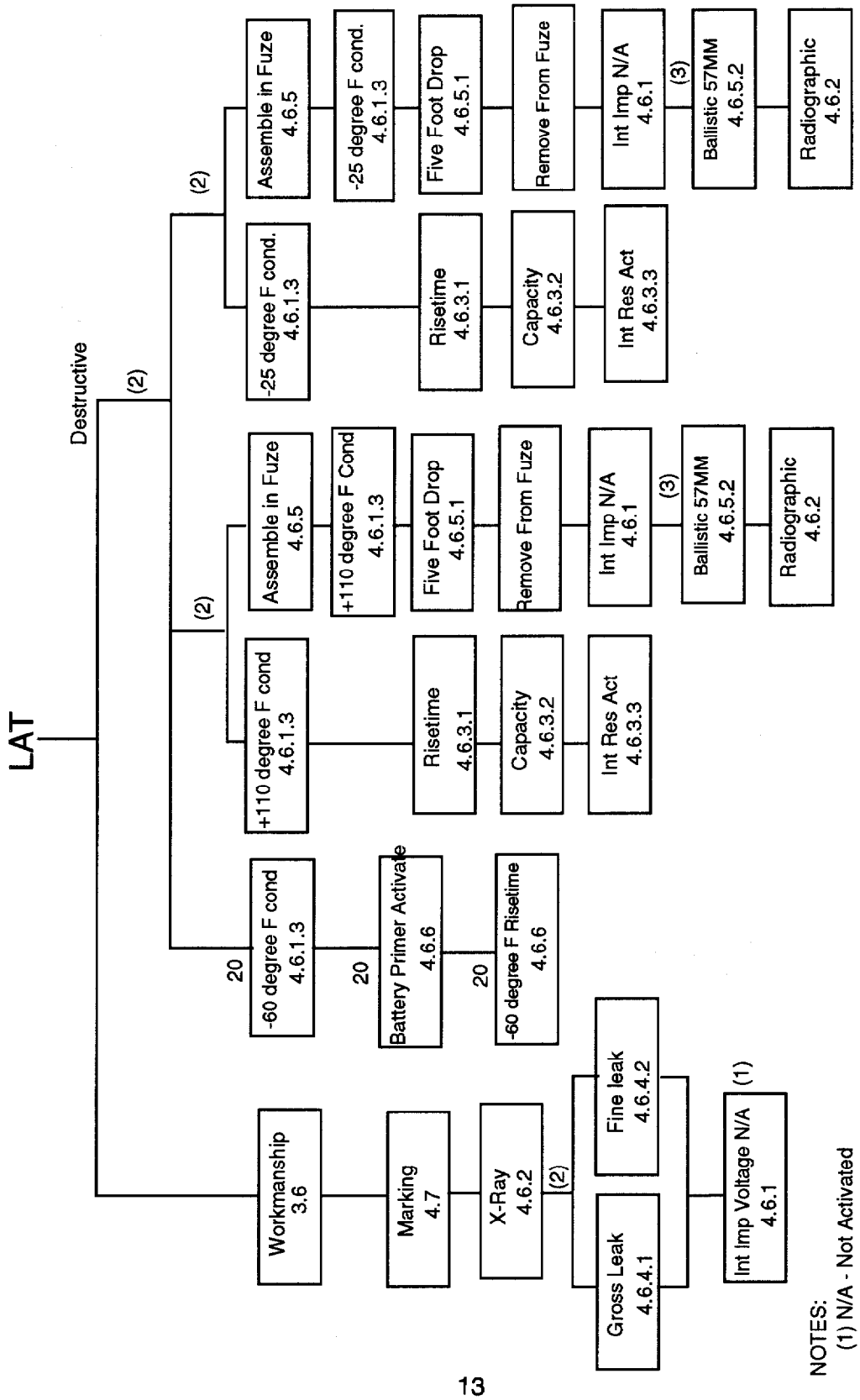
a. Parts bearing the same lot interfix number from one manufacturer.

b. Parts produced under continuous production at one manufacturing location, in accordance with stabilized methods or techniques.

4.4.1.2 Inspection and lot acceptance.. Inspection shall be as specified in 4.4.2.1, and Figure 2. Lots acceptance shall be in accordance with 4.4.2.1.

4.4.1.3 Individual unit inspections. Individual unit inspections shall be in accordance with 4.4.2.2 through 4.4.2.23.

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NOTES:
(1) N/A - Not Activated
(2) If N is not even, the odd battery shall be included in the left hand path of the flow.
(3) See paragraph 3.7.3

RESERVE BATTERY, M762/M767
LOT ACCEPTANCE FLOW

FIGURE 2

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4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

TABLE II Attributes sampling inspection

	Inspection Levels					
Lot Size	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	32	20	8
281 to 500	*	125	32	32	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13
150001 to 500000	1250	500	200	125	50	13
500001 and over	1250	500	200	125	50	13

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more failures/defects for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.6). When required, the contractor shall demonstrate that the

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effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.1	Battery, single cell, reserve	SHEET 1 OF 1	12550985
			NEXT HIGHER ASSEMBLY: None

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Shim, Ampule Support Missing	Level IV	3.5.1	Visual
102	Disc, Insulator Missing	Level IV	3.5.1	Visual
103	Ampule Support Pads Missing	Level IV	3.5.1	Visual
104	Weld Integrity	Level IV	3.5.1	Visual
105	Leak Test	Level IV	3.5.2	4.6.4.1
106	Non-activation Voltage and Resistance	Level IV	3.4.1	4.6.1
107	Risetime, Capacity, Internal Impedance Test	Level V	3.4.2.2	4.6.3
108	Battery Primer Activation	Note 2	3.4.2	4.6.6
109	Radiographic Examination	Level IV	3.9	4.6.2
110	Environmental Test		3.7.1	4.6.5.1
111	Physical Dimensions	Level IV	3.5.1	Dwg. 12550985
112	Ballistic Test	N=36,Note 1	3.7.3	4.6.5.2
Minor				
201	Max Diameter	Level V	3.5.1	Gage
202	Max Length	Level V	3.5.1	Gage
203	Marking	Level V	3.8	4.7
204	Evidence of Poor Workmanship	Level V	3.10	Visual

NOTES:
1. Performed by Gov't.
2. Sampling Plan shall be 20-3-4 & 20-0-1 at (-60 F only) if applicable (see 4.5.6).

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Disc, insulator	1 OF 1	12550986
			NEXT HIGHER ASSEMBLY: 12550985

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material Improper		3.5.1	Certification of Conformance
Minor				
201	Outside Diameter (Max)	Level V	3.5.1	Gage
202	Inside Diameter (Min)	Level V	3.5.1	Gage
203	True Position of ID with OD	Level V	3.5.1	Gage
204	Thickness Improper	Level V	3.5.1	Gage
205	Evidence of Poor Workmanship	Level V	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Pad, ampule support	1 OF 1	12550987
			NEXT HIGHER ASSEMBLY: 12550985

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material Improper		3.5.1	Certificate of Conformance
Minor				
201	Inside Diameter Improper	Level V	3.5.1	Gage
202	Outside Diameter Improper	Level V	3.5.1	Gage
203	Thickness	Level V	3.5.1	Gage
204	True Position of OD	Level V	3.5.1	Gage
205	Evidence of Poor Workmanship	Level V	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Shim, ampule support	1 OF 1	12550988
			NEXT HIGHER ASSEMBLY: 12550985

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material Improper		3.5.1	Certificate of Conformance
Minor				
201	True Positions of the 6 Diameters	Level V	3.5.1	Gage
202	Measurements of 6 Diameters	Level V	3.5.1	Gage
203	Total Height	Level V	3.5.1	Gage
204	Evidence of Poor Workmanship	Level V	3.10	Visual
205	Slot Height and Width	Level V	3.5.1	Gage

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Terminal plate assembly	1 OF 1	12550995
			NEXT HIGHER ASSEMBLY: 12550985

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Pin Protrusion inside	Level IV	3.5.1	Gage
102	Pin Protrusion outside	Level IV	3.5.1	Gage
Minor				
201	Pin True Position	Level V	3.5.1	Gage
202	Evidence of Poor Workmanship	Level V	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Plate, terminal	1 OF 1	12550996
			NEXT HIGHER ASSEMBLY: 12550995

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Overall Thickness	Level IV	3.5.1	Gage
Minor				
201	Outside Diameter Improper	Level V	3.5.1	Gage
202	Evidence of Poor Workmanship	Level V	3.10	Visual
203	OD True Position with I.D.	Level V	3.5.1	Gage
204	Taper not present	Level V	3.5.1	Gage

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Pin, terminal	1 OF 1	12550998
			NEXT HIGHER ASSEMBLY: 12550995

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material Improper		3.5.1	Certificate of Conformance
Minor				
201	Diameter Improper	Level V	3.5.1	Gage
202	Evidence of Poor Workmanship	Level V	3.10	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Insulator, bottom, inside	1 OF 1	12551007
			NEXT HIGHER ASSEMBLY: 12551025

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material Improper		3.5.1	Certificate of Conformance
Minor				
201	Diameter Improper	Level V	3.5.1	Gage
202	Thickness Improper	Level V	3.5.1	Gage
203	Evidence of Poor Workmanship	Level V	3.10	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-B-70824A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Ampule assembly, electrolyte	1 OF 1	12551015
			NEXT HIGHER ASSEMBLY: 12550985

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Overall Length (Max)	Level IV	3.5.1	Gage
102	Diameter (Max)	Level IV	3.5.1	Gage
103	Quantity of Electrolyte	Note 1	3.5.1	4.6.8.2
104	Electrolyte Material Improper		3.5.1	Certificate of Conformance
Minor				
201	Evidence of Poor Workmanship	Level V	3.10	Visual

NOTES:
1. Number of sample units shall be in accordance with paragraph 4.5.8.1.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Cell assembly	1 OF 1	12551020
			NEXT HIGHER ASSEMBLY: 12550985

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Open circuit voltage check	100%	3.1	Gage
Minor				
201	Evidence of Poor Workmanship	Level V	3.10	Visual
202	Opening of Anode Assembly Aligned with opening of Cathode	Level V	3.5.1	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.11	Case assembly, insulator	SHEET 1 OF 1	12551025
			NEXT HIGHER ASSEMBLY: 12551020

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor 201	Evidence of Poor Workmanship	Level V	3.10	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Insulator, bottom, outside	1 OF 1	12551027
			NEXT HIGHER ASSEMBLY: 12551025

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material Improper		3.5.1	Certification of Conformance
Minor				
201	Outside Diameter	Level V	3.5.1	Gage
202	Inside Diameter	Level V	3.5.1	Gage
203	True Position of ID with OD	Level V	5.3.1	Gage
204	True Positions of 8 slots	Level V	3.5.1	Gage
205	Evidence of Poor Workmanship	Level V	3.10	Visual
206	Overall Thickness	Level V	3.5.1	Gage
207	Slot Depths	Level V	3.5.1	Gage

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.13	Separator, bottom	SHEET 1 OF 1	12551028
			NEXT HIGHER ASSEMBLY: 12551025

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material Improper		3.5.1	Certification of Conformance
Minor				
201	Diameter Improper	Level V	3.5.1	Gage
202	Evidence of Poor Workmanship	Level V	3.10	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.14	Cell assembly, anode/cathode	SHEET 1 OF 1	112551029
			NEXT HIGHER ASSEMBLY: 12551020

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Vertical Separator Presence	Level V	3.5.1	Visual
Minor 201	Evidence of Poor Workmanship	Level V	3.10	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.15	Anode assembly	1 OF 1	12551030
			NEXT HIGHER ASSEMBLY: 12551029

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Anode Substrate weld seam Not Exposed in Diagonal Slot of Anode Assembly	Level IV	3.5.1	Visual
102	Anode Pad Penetrating Perforations in Anode Substrate	Level IV	3.5.1	Visual
103	Anode Pad Not Adhered to Anode Substrate on 100% of Anode Pad Surface	Level IV	3.5.1	Visual
104	Anode Pad within 0.005 from Top or bottom of Anode Substrate	Level V	3.5.1	Gage
105	Anode Pad Material		3.5.1	Certificate of Conformance
Minor				
201	Spiral Gap in Anode Pad	Level V	3.5.1	Gage

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.16	Substrate, anode, formed	SHEET 1 OF 1	12551032
			NEXT HIGHER ASSEMBLY 12551030

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Weld Integrity Improper	Level IV	3.5.1	Visual
Minor				
201	Inside Diameter Improper	Level V	3.5.1	Gage
202	Matchup of Lap	Level V	3.5.1	Visual
203	Evidence of Poor Workmanship	Level V	3.10	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.17	Substrate, anode	1 OF 1	12551033
			NEXT HIGHER ASSEMBLY: 12551032

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Material Improper		3.5.1	Certification of Conformance
Minor				
201	Length Improper A	Level V	3.5.1	Gage
202	Width Improper B	Level V	3.5.1	Gage
203	Bottom Border C	Level V	3.5.1	Gage
204	Screen Hole Pattern D	Level V	3.5.1	Gage
205	Step In Lead E	Level V	3.5.1	Gage
206	Evidence of Poor Workmanship	Level V	3.10	Visual
207	Overall Thickness F	Level V	3.5.1	Gage

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	Cathode assembly	1 OF 1	12551035
			NEXT HIGHER ASSEMBLY: 12551029

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Composition Cathode Not to Extend Beyond Edges of Cathode Substrate	Level IV	3.5.1	Visual
202	Assembly Thickness	Level V	3.5.1	Gage
203	Evidence of Poor Workmanship	Level V	3.10	Visual
204	Composition Cathode to Edge of Substrate Dimension	Level V	3.5.1	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-B-70824A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.19	Substrate, cathode	1 OF 1	12551036
			NEXT HIGHER ASSEMBLY: 12551035

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Crtical	None defined			
Major				
101	Material Improper		3.5.1	Certification of Conformance
Minor				
201	Length Improper A	Level V	3.5.1	Gage
202	Width Improper B	Level V	3.5.1	Gage
203	Bottom Border C	Level V	3.5.1	Gage
204	Screen Hole Pattern D	Level V	3.5.1	Gage
205	Evidence of Poor Workmanship	Level V	3.10	Visual
206	Step in Leads E	Level V	3.5.1	Gage
207	Overall Thickness F	Level V	3.5.1	Gage

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.20	Cathode Composition	1 OF 1	12551037
			NEXT HIGHER ASSEMBLY: 12551035

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Weight Improper	Level IV	3.5.1	Gage
102	Thickness Improper	Level IV	3.5.1	Gage
103	Material Improper	Level IV	3.5.1	Certification of Compliance
104	Percentage of Acetylene Black	Note 1.	3.5.1	Gage
105	Percentage of PTFE	Note 1.	3.5.1	Gage
Minor				
201	Length Improper	Level V	3.5.1	Gage
202	Width Improper	Level V	3.5.1	Gage
203	Evidence of Poor Workmanship	Level V	3.10	Visual

NOTES:
1. Record weight and calculate percentage of raw material prior to mix, data shall be retained for review by customer.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.21	Separator, vertical	1 OF 1	12551039
			NEXT HIGHER ASSEMBLY: 12551029

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Materials Improper		3.5.1	Certificate of Conformance
Minor				
201	Length Improper	Level V	3.5.1	Gage
202	Width Improper	Level V	3.5.1	Gage
203	Evidence of Poor Workmanship	Level V	3.10	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-B-70824A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.22	Case	1 OF 1	9275592
			NEXT HIGHER ASSEMBLY: 12551025

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Materials Improper		3.5.1	Certificate of Conformance
102	Hardness Improper	Level IV	3.5.1	3.6
103	Inside Diameter Improper	Level IV	3.5.1	Gage
104	Length Improper	Level IV	3.5.1	Gage
105	Thickness Improper	Level IV	3.5.1	Gage
Minor				
201	Finish Improper	Level V	3.5.1	Visual
202	Evidence of Poor Workmanship	Level V	3.10	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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CLASSIFICATION OF CHARACTERISTICS

MIL-B-70824A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.23	Ampule blank	1 OF 1	12551016
			NEXT HIGHER ASSEMBLY: 12551015

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Ampule Diameter Improper	Level IV	3.5.1	Gage
102	Ampule Length to Shoulder Improper	Level IV	3.5.1	Gage
103	Stricture OD Improper (2 places)	Level IV	3.5.1	Gage
104	Stricture ID Improper	Level IV	3.5.1	Gage
105	Stricture True Position Improper	Level IV	3.5.1	Gage
106	Wall Thickness Improper	Level IV	3.5.1	Gage
107	Bottom Thickness Improper	Level IV	3.5.1	Gage
108	Material Improper	Level IV	3.5.1	Certificate of Conformance
Minor				
201	Overall Length	Level V	3.5.1	Gage
202	Evidence of Poor Workmanship	Level V	3.10	Visual

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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4.5 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. See Section 6 of the MIL-A-48078 and 6.3 herein.

4.6 Methods of inspection.

4.6.1 Non-activation impedance and voltage.

4.6.1.1 Voltage (Non-activated). Batteries in this specification shall be measured for non-activation voltage using an acceptable digital millivolt meter, as approved by the Government having a minimum input impedance of 1E+6 ohms with 500K ohms across the cell. Any battery which measures more than five (5.0) millivolts shall be classed defective and removed from further tests.

4.6.1.2 Internal impedance (Non-activated). The internal impedance of a non-activated battery shall be measured with a digital volt-ohm meter or other suitable instrument. The non-activated batteries built to this specification shall have a measured impedance of not less than one megohm.

4.6.1.3 Preconditioning. Depending on the test flows of Figures 1A, 1B or Figure 2, the non-activated voltage and internal impedance tests may be preceded by temperature conditioning. Where called for, the batteries shall be conditioned to the test temperature specified in Figures 1A, 1B or Figure 2 for a minimum of two hours and held to a temperature accuracy of ± 5 degrees F. Testing shall be performed within 2 minutes after removal from the conditioning chamber.

4.6.2 Radiographic examination. Inspection equipment and image quality of the exposed x-ray films for the examination shall be satisfactory to the purchaser. The direction of the x-ray beam employed in the inspection shall be perpendicular to the cylindrical wall of the battery. Retain x-ray films in the file during production. (Non-destructive test).

X-ray negatives of the batteries shall be examined for the following:

a. There shall be continuity of the anode lead.

b. There shall be no sharp bends in the anode lead.

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c. The anode lead shall not be bent against the outside case.

d. There shall be no discernible sharp bends, blocking or damage to the cathode leads.

e. There shall be no discernible damage to the anode or cathode screens.

4.6.3 Risetime, capacity and impedance test. Batteries to be tested shall be preconditioned prior to activation for a minimum of two hours at the test temperatures indicated in the FAAT or Lot Acceptance Test flows of Figures 1A, 1B or 2, per 4.6.1.3. The test temperature shall be maintained within ± 5 degrees F. The batteries shall remain at the test temperatures for the duration of testing. For activation and risetime tests, the batteries may be removed from the temperature chambers for a period not exceeding two minutes.

4.6.3.1 Risetime test. Following conditioning, each battery to be tested for risetime shall be connected to a 13K ohm ± 1% load prior to activation. Activation of the battery for this test shall be achieved per procedure 12550752, Risetime, Capacity and Internal Impedance Test for M762 Reserve Cell, Battery Primer Activation. Time shall be measured from immediately after the activation shock until the terminal voltage across the 13K ohm load reaches 2.30 volts. The risetime shall be measured with a storage oscilloscope or other suitable instrument and recorded. The measured risetime shall not exceed the values shown below for the associated temperature. The test shall be performed at the conditioning temperature.

The maximum measured risetimes shall not exceed the following values:

TEMPERATURE	MAXIMUM RISE TIME TO 2.3 VOLTS
+110 degrees F	300 Milliseconds
Ambient	800 Milliseconds
-25 degrees F	1500 Milliseconds
-60 degrees F	2500 Milliseconds

4.6.3.2 Capacity test. Those batteries tested for rise time per 4.6.3.1 above shall be used for capacity testing. The batteries shall be maintained at the conditioning temperature for 360 + 4, -0 hours following activation with the 13K ohm

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resistive loads attached. The terminal voltage, time, and temperature shall be recorded every 4 hours for the 360 hour period. The terminal voltage shall not fall below the following values for each temperature:

TEMPERATURE	MINIMUM VOLTAGE (RL = 13K)
+110 degrees F	3.50 Volts
Ambient	3.40 Volts
-25 degrees F	2.90 Volts
-60 degrees F	2.40 Volts

4.6.3.3 Internal impedance (Activated). The batteries that were previously tested for capacity 4.6.3.2 shall be tested for internal impedance at the completion of the 360 hour +4, -0 period. A 150 ohm ± 2% resistive load shall be applied for 50 ms ± 1 ms in parallel with 13K ohm ± 1% load for each battery. Minimum battery terminal voltage under load shall be measured and also measured at 5 ms after removal of the load (recovery voltage). The recovery voltage shall be a minimum value of 2.8 for all temperatures. The minimum terminal voltage under load conditions shall be:

TEMPERATURE	TERMINAL VOLTAGE
+110 degrees F	3.10 Volts
Ambient	3.00 Volts
-25 degrees F	2.20 Volts

4.6.4 Leak test.

4.6.4.1 Gross leak. Each battery shall be subjected to the gross leak test. The test shall be performed in accordance with MIL-STD 202, Method 112, Condition B. (Any evidence of bubbles shall be cause for rejecting the gross leak test using the fine leak test equipment of 4.6.4.2, provided the equipment has the capability of gross leak detection). Any battery failing to meet the leak rate requirements shall be rejected.

4.6.4.2 Fine leak. Batteries manufactured to this specification shall be tested in accordance with ASTM-E-493, Method A. The volume to be used for obtaining the maximum leak rate shall be 0.6cc. The maximum leak rate shall be less than 5×10^{-7} cc per second (helium). Any battery failing this test shall be rejected.

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4.6.5 Environmental and ballistic testing.

4.6.5.1 1.5 Meter (5 foot) drop test. Batteries to be drop tested shall be preconditioned prior to drop testing at the temperatures shown in Figures 1A or Figure 2. The preconditioning temperatures shall be maintained at ± 5 degrees F for a minimum of 4.0 hours. The batteries shall be assembled into M762 fuze assemblies prior to temperature conditions. The assemblies need not be operational, but shall be mechanically representative of the fuze.

After temperature conditioning the fuze assemblies shall be subjected to a 1.5 meter drop test in accordance with MIL-STD-331, Test 111.1, Procedure 2, Condition A with the exception that less than 10 batteries may be dropped. The elapsed time between removal from the conditioning chamber and the start of the drop test shall not exceed 15 minutes. The batteries shall be monitored for 24 hours following the drop test. To assure that no battery was activated by this test, monitoring shall be performed with a voltage recording system with a minimum input impedance of 1E+6 ohms. A 500K ohm ± 5% resistive load shall be across the battery during monitoring. Any battery which measures greater than 5 millivolts (5.0 MV) shall be failed and considered activated.

4.6.5.2 Ballistic testing. The test is performed by the Government on batteries that are activated at ambient per 4.6.6. Batteries shall meet the risetime requirement per 4.6.6.

Following activation the batteries shall be mounted in a T-can projectile (inert) with a weight of 6.36 pounds. The projectile shall be exposed for a minimum of four hours at the conditioning temperatures specified in Figure 1B or Figure 2.

The projectile shall contain circuitry to continuously monitor battery terminal voltage for 50 milliseconds ± 1.0 millisecond after setback and detect voltage interruptions greater than 5.0 microseconds ± 10%.

The projectile shall be fired in a vertical recovery test with a 57MM type M1 weapon. The propellant shall be type MP-MI (RAD 68308), with a nominal 41-ounce quantity. The quantity may be adjusted according to ambient conditions.

A voltage interruption of more than 5.0 ± 10% microseconds during the first 50 milliseconds following setback shall be considered a battery failure.

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4.6.6 Battery primer activation. Batteries to be activated using battery primers shall be tested according to paragraphs (4.0, and 5.1 through 5.5),of drawing 12550752. If any unit exceeds the 5000 milliseconds risetime or the (pass/fail)of 0/1 for the operating limits specified , the lot shall be rejected. If exactly four (4) units exceed the required maximum risetime (2500 milliseconds), retest per 4.6.6.1. If more than four (4) units in the sample of 20 exceed the required maximum risetime, reject the lot.

The maximum measured risetimes shall not exceed the following values:

TEMPERATURE	MAXIMUM RISETIME TO 2.3 VOLTS
+110 degrees F	300 Milliseconds
Ambient	800 Milliseconds
-25 degrees F	1500 Milliseconds
-60 degrees F	2500 Milliseconds

4.6.6.1 Battery primer activation retest. If a second sample is required, testing shall be per 4.6.6 except for the following; battery only shall be conditioned to -60 degrees F. A sample of 20 units shall be tested and if any unit exceeds the required maximum risetime the lot shall be rejected.

4.6.7 Ampule heat soak. Each ampule assembly (12551015) shall be placed in a chamber heated to 165 degrees F ± 5 degrees F for a minimum of 24 hours. After the soak the ampules shall be removed from the chamber and visually examined under 10X magnification for signs of leakage, cracks, or capillary passage in the ampule nipple. Any unit with these defects shall be classed defective and removed from the lot. (Non-destructive test).

4.6.8 Electrolyte fill. The ampules shall be filled with .53cc - 0.04cc of Thionyl Chloride Electrolyte, specification MIL-T-63420.

4.6.8.1 Electrolyte volume. At the start of each day, ten electrolyte samples shall be taken and the volume of electrolyte checked. Do not proceed with production until all ten samples are accepted. At the end of each two-hour period during the continuous production run and at the end of each day's production, five units shall be randomly filled and the volume of electrolyte checked. If any sample fails to contain the proper amount of electrolyte, that quantity of ampule assemblies filled since the last acceptable check shall be rejected. This

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check shall be made in accordance with 4.6.8.2.

4.6.8.2 Electrolyte check. Individually pre-weigh and record the weight of ten ampule blanks. After filling, reweigh the filled ampule blanks and calculate the weight of the electrolyte. Using the specific gravity of the electrolyte batch, calculate the volume. Repeat at intervals specified in 4.6.8.1 with five blanks.

4.6.9 Die checks for cast, molded, and stamped parts.

4.6.9.1 Die check for cast and molded parts. Samples of at least three parts from each cavity shall be inspected in accordance with the applicable Classification of Characteristics and Tests paragraph to qualify a new or reworked cavity for use in production. Individual cavity identification must be provided. If any defective parts are found during qualification of the cavity due to a defective cavity, the cavity producing the defective part shall not be used in production. All cavities formerly removed from production because of a fault may, after reworking, be returned to production, providing they pass the qualification test. In addition, a minimum sample of one part from each cavity shall be inspected for each lot acceptance in accordance with the applicable Classifications of Characteristics Table. If the one piece sample passes the inspection, the contractor may proceed with the lot. This, however, does not relieve the contractor from ensuring that the entire lot is acceptable. If the one piece inspected fails the inspection of the Classification of Characteristics specified, the lot shall be rejected.

4.6.9.2 Die check for stamped parts. A sample of at least three parts from each die shall be inspected in accordance with the applicable Classification of Characteristics (see section 4.4). If any defective parts are found during qualification of the tool due to a defective tool, the tool producing the defective part shall not be used in production. All tools removed from production because of some fault may, after reworking, be returned to production, providing they pass the qualification test above. In addition, a minimum sample of one part from each cavity shall be inspected for each lot acceptance in accordance with the applicable Classifications of Characteristics Table. If the one piece sample passes the inspection, the contractor may proceed with the lot. This, however, does not relieve the contractor from ensuring that the entire lot is acceptable. If the one piece inspected fails the inspection of the Classification of Characteristics specified, the lot shall be rejected.

4.6 Marking. See Drawing 12550985. Batteries procured to

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this specification shall be marked with the lot/date code. In addition, the minimum protective packaging shall show the part number and be marked with "WARNING: EXCESSIVE HEATING OF THE BATTERY MAY CAUSE A RUNAWAY REACTION CAUSING THE CELL TO OPEN MILDLY (VENT) OR VIOLENTLY (EXPLOSION). DO NOT INCINERATE, SHORT CIRCUIT, USE OR STORE OUTSIDE SPECIFIED TEMPERATURE RANGE, PUNCTURE, CRUSH, DISASSEMBLE, RECHARGE OR FORCE OVER-DISCHARGE (REVERSE)."

5. PACKAGING

5.1 Packaging and preservation. Battery assemblies purchased to this specification shall be packed in accordance with MIL-STD-1169. The batteries must be packaged such that shorting of a cell cannot occur.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The battery assembly covered by the specification is intended for use in munitions assemblies.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1)

c. Requirements for submission of first article sample.

6.3 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAR-I, Picatinny Arsenal, New Jersey 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

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6.4 Distribution of ammunition data cards, inspection and test results, process controls. Distribution of ammunition data cards, inspection and test results, process controls shall be sent to:

Commander
ARDEC
ATTN: SMCAR-QAR-I
Picatinny Arsenal, New Jersey 07806-5000.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development, and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.6 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.7 Subject term (key word) listing.

Activation
Dry Room control
Electrolyte
Internal
Impedance
Rise Time

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 6135-A229)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-F-70824A (AR)	940602

3. DOCUMENT TITLE: BATTERY, LITHIUM THIONYL CHLORIDE, SINGLE CELL RESERVE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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